

# Anoop Krishna

## AI/ML Engineer

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### Professional Summary

AI/ML Engineer who builds end-to-end — from agentic AI pipelines and real-time computer vision systems to production RAG architectures. Hands-on experience integrating LLMs (GPT-4, Claude, LLaMA) into customer-facing products at Rubixe, with a project portfolio spanning multi-agent systems, ADAS safety pipelines, and hybrid retrieval engines. Backed by 6 years of international professional experience in finance and operations across Saudi Arabia and UAE — bringing analytical rigor, cross-functional ownership, and the discipline to ship. Currently building QuickKit, a live monetized web app on Vercel.

### Experience

#### Data Scientist (Intern) | Rubixe, Bengaluru

Jan-2024 – Aug-2024

- Built a GenAI-powered software architecture generator using GPT-4 and Claude APIs, cutting project delivery time by 30%.
- Integrated GPT, Claude, and Mistral LLMs into customer-facing workflows using LangChain, improving satisfaction scores by 22%.
- Developed and deployed predictive ML models for product delivery forecasting using Scikit-learn and Python.
- Led internal R&D on emerging AI tools including LangChain, vector databases, and prompt engineering techniques, integrating findings into production pipelines.
- Conducted peer code reviews for ML codebases, enforcing best practices around model evaluation, API integration, and code efficiency.

#### Store Supervisor | Abu Dhabi, UAE

Feb-2021 – Aug-2023

- Managed end-to-end store operations, inventory tracking, and vendor coordination across a high-volume retail environment.
- Streamlined inventory processes through data-driven analysis, reducing stock discrepancies and improving supply chain efficiency.

#### Accountant | Riyadh, Saudi Arabia

Oct-2017 – March-2020

- Handled financial reporting, ledger management, and reconciliation across multi-currency transaction environments.
- Developed analytical workflows for expense tracking and reporting that improved audit readiness and accuracy.

### PROJECTS

#### Production RAG Pipeline with Automated Evaluation

[github.com/rag](https://github.com/rag)

- Built hybrid RAG system (FAISS + BM25) using LangChain and Groq LLaMA 3.3 70B, improving retrieval accuracy by 40% over pure semantic search.
- Automated evaluation with RAGAS across 4 metrics, achieving 0.87+ average score on Faithfulness, Relevancy, Precision, and Recall.
- Integrated LangSmith tracing for end-to-end observability, capturing latency, token usage, and chain breakdown per query.
- Deployed Streamlit dashboard with live Q&A, PDF upload, and evaluation runner — zero API cost using local HuggingFace embeddings.
- Engineered modular architecture with independent ingestion, retrieval, generation, and evaluation layers — each independently testable and production-ready.

#### Real-Time Driver Safety System (ADAS)

[github.com/adas](https://github.com/adas)

- Built real-time CV pipeline detecting drowsiness, distraction, and seatbelt violations using YOLOv8 and MediaPipe on live webcam feeds.
- Achieved 90%+ detection accuracy with sub-200ms inference latency using optimized OpenCV and PyTorch frame processing.
- Designed 3 independent detection modules — eye aspect ratio, head pose estimation, and object detection — in a unified pipeline.
- Deployed Streamlit dashboard with real-time alert overlays and exportable safety reports targeting ADAS applications.
- Designed system architecture with extensibility for edge deployment in mind, with ONNX-compatible model export for real-world automotive hardware integration.

## Multi-Agent Research System

[github.com/agent](https://github.com/agent)

- Built a multi-agent research pipeline using LangGraph and Groq LLaMA 3.3 70B, decomposing complex queries into specialized sub-tasks across coordinated AI agents.
- Integrated Tavily search API for real-time web retrieval, delivering grounded, citation-backed research reports in under 30 seconds.
- Designed stateful agent graph with LangGraph managing agent transitions, memory, and tool calls across the full research workflow.
- Deployed as a Streamlit app with configurable agent depth, structured report output, and LangSmith tracing for full pipeline observability.
- Implemented configurable agent depth and dynamic tool routing, enabling the system to handle both shallow factual queries and deep multi-step research tasks.

## Skills

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- **Core:** Python · SQL · Git · PyTorch · TensorFlow · Scikit-learn · Pandas · NumPy
- **GenAI & Agents:** LangChain · LangGraph · RAG (FAISS + BM25) · RAGAS · Groq · OpenAI API · Claude API · Mistral · LangSmith · HuggingFace
- **Computer Vision:** YOLOv8 · MediaPipe · OpenCV · CNNs · ONNX · Real-Time Inference
- **Deployment & Frontend:** Docker · FastAPI · Streamlit · React · React Native · Vite · Tailwind CSS

## Education

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**B.Sc. Electronics with Computer Science**

May-2013 – Aug-2016

University of Calicut

Coursework: Database Management Systems, Probability, Statistics

## Certification

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- **Certified Data Scientist- DataMites**
- **Certified Data Scientist – IABAC (International Association for Business Analytics Certification)**

## Currently Building

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- **QuickKit**  
A browser-based utility web app with 12+ tools across Dev Tools, Text Tools, and Finance categories. Built with React, Vite, and Tailwind CSS — live on Vercel with Google AdSense monetization. [QuickKit.Dev](#)
- **Parsely**  
AI-powered food label scanner mobile app built with React Native and Expo. Scans product barcodes/labels, analyzes ingredients against a custom user health profile, and tracks nutritional analytics — acting as a personalized health companion.